

PHARMACOKINETICS EXTRA PRACTICE PROBLEMS

STUDENTS VERSION

1

PHARMACOKINETIC REVIEW

Please make sure you have a printout of the PEBC MCQ reference sheet from the following website before the review lecture:

<https://pebc.ca/wp-content/uploads/pdfs/Qual%20EN/FormulasQual.pdf>

2

John McCarthy is a 60 kg patient who is started on an IV infusion of amikacin at a rate of 5mg/min. Assume a volume of distribution of 30 L, a half-life of 1 hr, and a clearance of 20 L/h. What would be the plasma concentration of amikacin at steady state?

3

MR is a 30-year-old female who weighs 55 kg and has been prescribed a single dose Drug Y at 20 mg/kg. Drug Y has an elimination half-life of 5 hours, a volume of distribution of 0.25 L/kg, and is 30% protein bound. Drug Y follows a one-compartment model and first-order kinetics.

- Calculate the initial plasma concentration (C_{p0}) of Drug Y
- Predict the plasma concentration of Drug Y after 10 hours
- Calculate how much of the drug remains in the body after 10 hours

4

Maria is 57-year-old female who weighs 145 pounds. She was admitted to the hospital with a severe UTI and was given an IV dose of Cephalexin 500mg. Two hours later, a blood sample was withdrawn to measure the drug concentration. If all the medication was present in the body and not eliminated during the 2 hours, what is the expected peak plasma concentration following the single Cephalexin dose? Assume the volume of distribution to be 0.2 L/Kg and the salt factor to be 1.

5

Elvis Pieer is 68 kg patient who is started on an IV infusion of a new antibiotic. If the target plasma concentration for this patient is 4mg/L, what would be the pharmacokinetic parameters for this antibiotic given that its volume of distribution = 30 L, half-life is 17 hr, and clearance is 2 L/h?

6

Mr. Anderson was infused with a medication (Med X) for a duration of 6 hours at a rate of 2 mg/hr. The medication has a volume of distribution (V_d) of 10 L and an elimination rate constant (k) of 0.02 hr^{-1} . What is the concentration of Med X in Mr. Anderson's body 2 hours after the infusion was stopped?

7

Samir Murad, is a 45-year-old male patient with a history of chronic pain who was administered an intravenous (IV) bolus dose of a medication. Ten hours post-administration, the drug's plasma concentration was measured at 4 mg/L. Given a known volume of distribution (V_d) of 10 L and an elimination half-life ($t_{1/2}$) of 5 hours, calculate the dose administered to the patient.

8

A 30-year-old female presents to the emergency department complaining of severe dysmenorrhea. The physician decides to administer a loading dose of an analgesic medication to promptly alleviate her pain. The analgesic administered has the following pharmacokinetic characteristics: a half-life of 10 hours, a clearance rate of 0.04 L/min, and a volume of distribution of 40 L. To achieve complete pain relief, the target plasma concentration for the drug has been set at 4 mcg/mL. Calculate the loading dose needed for this patient to ensure rapid pain relief.

9

What is the necessary infusion rate for aminophylline considering that the desired steady-state plasma concentration is 15 mg/L? The average half-life is about 4 hrs and the apparent volume of distribution is about 25 liters.

10

Mr. Jackson has been suffering from severe infection in his right foot. The doctor decided to give him a single dose of 220 mg of an antibiotic in the form of an infusion that lasts 60 min. The plasma concentration after one hour of the infusion is 14.4 mg/L. How many hours after the end of infusion will the concentration reach 2mg/L, considering that the $K_{el}=0.35 \text{ hr}^{-1}$? What is the half life of this antibiotic?

11

In a phase II clinical trial, a 20-mg dose of a newly developed medication exhibiting first-order elimination kinetics was administered intravenously to healthy subjects. The drug's volume of distribution (V_d) was determined to be 50 liters. If a 40-mg dose of the same medication is administered, what is the calculated volume of distribution for this higher dose?

12

John, a 60-year-old patient who weighs 75 kg, is receiving an intermittent IV infusion of Drug Y at a dose of 300 mg. After the administration of Drug Y, John's peak plasma concentration is measured and found to be 26 mg/L. The elimination rate constant for Drug Y is determined to be 0.2 h^{-1} and volume of distribution is 0.8 L/Kg . Calculate the total clearance (ml/min) of Drug Y in John.

13

A patient received a new antibiotic at a dosage of 10 mg/Kg twice daily, at 09:00 and 21:00, administered as a 45-minute intravenous infusion. The peak concentration was evaluated at 11:00 and measured as 40 mcg/dL . The trough level was measured 1 hour before the subsequent dose, and it was observed to be 3 mcg/dL . What is the drug's half-life ($t_{1/2}$)?

14

Julia Roberts is a 32-year-old female who weighs 150 pounds needs an intravenous infusion of antibiotic (X). The physician would like to achieve a steady state concentration (C_{ss}) of 15 mg/dl by infusing the dose over 8 hours without giving a loading dose. Drug (X) has an elimination $t_{1/2}$ of 1.4 hours and a volume of distribution (VD) of 8L.

- A) What is the rate of intravenous infusion that should be recommended for this patient?
- B) How long after starting the IV infusion would the plasma concentration reach 95% of the desired steady state concentration (C_{ss})?
- C) What is the recommended loading dose?
- D) What is the total clearance rate of this antibiotic in this patient in mL/min?
- E) If the patient's renal clearance of this antibiotic is 40ml/min, what is the hepatic clearance?

15

Sarah Johnson, a 30-year-old female, was administered a dose of Drug (Y), and two hours later, her serum drug concentration was found to be 43.2 mg/L. Ten hours after the dose was given, her serum drug concentration was measured again and found to be 12 mg/L. Assuming first-order kinetics for the drug, what is the half-life of drug (Y)?

16

William Brown, a 50-year-old male, who weighs 60 kg and has a history of heart failure, has been admitted to the hospital with worsening symptoms. The physician wants to start him on a new medication, Medication (X), and achieve a target concentration of 2.5 ng/mL with a loading dose. The medication has an elimination half-life of 3 hours and a volume of distribution (V_d) of 10 L/ kg. If the medication is available in a 100 mcg/mL solution with a bioavailability of 0.8 and a salt factor of 2, what would be the appropriate loading dose for William?

17

In a clinical setting, a male patient of 50 years of age was administered a single intravenous dose of 10 mg of an antibiotic. The serum drug concentration was subsequently assessed and found to be 75 mcg/dL. Given a patient weight of 180 pounds, what is the volume of distribution (V_d) in L/kg of the administered antibiotic?

18

What is the elimination half-life ($t_{1/2}$) for a drug that was administered intravenously, assuming the drug follows first-order kinetics and has complete distribution? The maximum serum drug concentration (C_{max}) was found to be 12 mcg/mL, and 24 hours later, the serum concentration had decreased to 1mcg/mL.

19

In order to achieve a target blood concentration (C_p) of 3.0 micrograms/L, a loading dose of Drug (B) is required for J.L., a patient with a body weight of 80 kg. Given a volume of distribution (VD) per Kg of body weight of 4 L, a salt factor (S) of 2, and a bioavailability (F) of 0.9, calculate the loading dose of the drug.

20

In the context of an IV infusion of a hypnotic agent with an infusion rate of 2 mg/hr, an elimination rate constant of 0.1 hr^{-1} , and a volume of distribution (one compartment) of 10 L, what is the recommended loading dose to achieve an immediate drug level of $2 \mu\text{g/mL}$?

21

Will Smith was infused with a hypnotic agent ($k = 0.01 \text{ hr}^{-1}$; $V_D = 10 \text{ L}$) at a rate of 2 mg/hr for 6 hours. What is the concentration of the hypnotic agent in John's body 2 hours after cessation of the infusion?

22

After administration of an antibiotic to patient K.S, the plasma drug concentration was measured at 4 $\mu\text{g/mL}$. If the clearance of the antibiotic is 20 mL/min, what is the elimination rate for the antibiotic?

23

A patient named MK is being treated with vancomycin IV 1g/12h. The therapeutic range for vancomycin peak concentration is 15-20 mg/L and the trough concentration is 10-15 mg/L. The patient's peak concentration after the fourth dose is 18 mg/L and the trough concentration immediately before the fifth dose is 11 mg/L. Assuming first-order elimination kinetics, what is the half-life ($T_{1/2}$) of vancomycin in this patient?

24

Maria Mounir, a 55-year-old patient with a body weight of 80 kg, was admitted to the hospital with a bloodstream infection. She was started on daptomycin therapy with a 2-gram dose administered as a 3-hour infusion. At 3 hours post-infusion, her plasma concentration of daptomycin was measured to be 50 mg/L, and at 10 hours post-infusion, it was measured to be 20 mg/L. Calculate the volume of distribution (Vd) of daptomycin for Maria.

25

A 70-year-old man with a body weight of 85 kg was given a new Ceftaroline antibiotic at a dose of 7 mg/kg via a single i.v. bolus injection. The antibiotic follows a one-compartment model in pharmacokinetics, with an elimination half-life of 3 hours. The drug has an apparent volume of distribution of 0.22 L/kg, and 25% of it is bound to plasma protein. What is the initial plasma drug concentration in this patient? What is the predicted plasma drug concentration at 10 hours after the dose? How much drug remains in the patient's body 8 hours after the dose?

26

Mike Johnson is a 42-year-old man with a weight of 62 kg presented to the clinic. His serum creatinine concentration was measured to be 1.0 mg/dl. The physician wants to calculate the patient's creatinine clearance.

27

A 25-year-old woman named Sarah Smith weighing 65 kg, presented to the clinic. The patient's serum creatinine concentration was measured to be 1.2 mg/dl. The physician wants to calculate the patient's creatinine clearance.

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Tobramycin 240 mg injection given by IV and plasma concentrations were determined as follows–

Time After Injection (hour)	Concentration (mg/lit)
1	21.6
2	17.8
3	14.8

What is the initial plasma concentration of the drug?